

RESEARCH ARTICLE | AUGUST 30 2011

One-dimensional planar hydrodynamic theory of shock ignition

R. Nora; R. Betti



Phys. Plasmas 18, 082710 (2011)

<https://doi.org/10.1063/1.3619827>



Articles You May Be Interested In

Shock ignition target design for inertial fusion energy

Phys. Plasmas (April 2010)

Influence of laser induced hot electrons on the threshold for shock ignition of fusion reactions

Phys. Plasmas (July 2016)

Strong shock generation by fast electron energy deposition

Phys. Plasmas (December 2013)

One-dimensional planar hydrodynamic theory of shock ignition

R. Nora and R. Betti^{a)}

Fusion Science Center and Laboratory for Laser Energetics, Physics and Astronomy, University of Rochester, Rochester, New York 14623, USA

(Received 7 April 2011; accepted 12 July 2011; published online 30 August 2011)

A one-dimensional planar compressible-piston-like model is used to investigate the basic physics behind shock-ignition inertial confinement fusion implosions. We discuss the theoretical limit set by rarefaction waves on the maximum hot-spot pressure achievable through conventional compression. Three ignitor shock techniques are presented to mitigate the effects of rarefaction waves, enhance the stagnation hot-spot pressure, and improve the ignition conditions. Elimination of rarefaction waves can lead to an $\sim 80\%$ increase in peak implosion pressures, while implosions augmented with ignitor shocks are shown to increase the peak pressures by a factor of ~ 4 . These techniques are then discussed and the optimal energy ratio between the initial shell kinetic energy and the ignitor pulse energy is given. © 2011 American Institute of Physics. [doi:10.1063/1.3619827]

I. INTRODUCTION

Recent theoretical^{1–9} and experimental results^{10,11} have indicated that launching a spherically convergent shock wave at the end of the acceleration phase improves the ignition conditions for inertial confinement fusion (ICF) implosions. Such an ICF scheme with a late shock is referred to as “shock ignition (SI).” SI is a scheme that is based on the principles of conventional central hot-spot ignition and uses a shock to augment the compression of the central hot spot above the ignition threshold. It was shown in Ref. 1 that a shock launched at the end of the main laser pulse can significantly enhance the hot-spot pressure and decrease the energy required for ignition by a factor $\Phi \approx (p^{\text{shock}}/p^{\text{no shock}})^{2-3}$, where p^{shock} is the hot-spot pressure enhanced by the ignitor shocks and $p^{\text{no shock}}$ is the hot-spot pressure without ignitor shocks. It was also shown in Ref. 1 that SI can lead to a pressure increase of about 70% in simulations of realistic ICF targets, therefore, reducing the energy required for ignition by about three- to five-fold. Analytical and simulation results by the NRL (Refs. 3 and 7) and the CELIA (Refs. 5, 8, and 9) groups were used to optimize the late shock timing and intensity. They carried out numerical simulations of realistic targets and found that such late shock leads to a significant hot-spot pressure enhancement and improvements of the ignition conditions. High gains at low implosion velocities were found by Perkins *et al.*⁴ using a late shock for laser pulses designed for the National Ignition Facility.¹² It is important to emphasize that such an ignition energy reduction applies to hydrodynamic-equivalent implosions, with the same implosion velocity, adiabat, laser intensity, and in-flight aspect ratio (IFAR).

The ignitor shock can be launched by a spike in the laser power or by energetic particles. Hydrodynamic drive is the conventional process for driving a shock through a spike in the laser power. The laser energy deposited during the power

spike drives a thermal wave toward the ablator surface, resulting in higher ablation near the capsule surface and stronger ram pressure. Similarly, a particle-driven shock can be generated by a moderately high intensity laser pulse ($>10^{16}$ W/cm²) used to accelerate electrons in the coronal plasma. If a burst of energetic particles is used, the temperature and pressure of the shell increase where the particle energy is deposited and the resulting high pressure drives the shock. In principle, laser-driven hot electrons can be very efficient in driving the ignitor shock if directed to the target and their energy is such that their stopping range is small compared to the target thickness at the time of the shock launching.¹³ If the energetic particles are produced by a high-intensity laser pulse, it may be possible to develop a mixed shock with a hydrodynamic as well as a particle drive. Initial particle-in-cell simulations indicate that the hot-electron energy is rather low (about 50 keV) and within the appropriate range for enhancing the shock strength.¹⁴

In central hot-spot ignition, the thermonuclear instability is triggered in the central hot spot. The hot spot is the relatively low-density central region of the spherical capsule that is heated to a temperature of several keV's by the PdV work of the surrounding dense shell. As the hot-spot pressure increases, the shell slows down from its peak velocity (i.e., the implosion velocity) reached during the acceleration phase and transfers a portion of its kinetic energy to the hot spot in the form of internal energy (i.e., pressure). Thick, massive shells can produce high-energy gains as a result of the large amounts of thermonuclear fuel. Because of their large mass, thick shells are driven at low implosion velocities and require very large drivers to trigger conventional hot-spot ignition. Indeed, the laser energy required for conventional hot-spot ignition is a strong function of the implosion velocity $E_L^{\text{ign}} \sim V^{-6.6}$ (Refs. 15 and 16), indicating the need for fast targets to minimize the required ignition driver energy. Fast targets are light and thin, however, and exhibit a large IFAR. The performance of fast targets is severely degraded by the growth of hydrodynamic instabilities that can break up the target while in flight. Therefore,

^{a)}Also at Department of Mechanical Engineering, University of Rochester, New York 14623, USA.

conventional ICF targets are designed to strike a balance between raising the implosion velocity while keeping the IFAR as low as possible.

Thick shells are hydrodynamically more stable and produce higher gains (if ignited) than their counterpart thin shells. The challenge with thick shells is their inefficiency in hot-spot compression to trigger the ignition process. This occurs for two reasons: thick shells take up more volume and, therefore, a larger fraction of the final internal energy goes into the shell rather than into the hot spot, and the finite compressibility of the shell causes different parts of the thick shell to transfer energy to the hot spot at different times. On the contrary, thin shells are very effective in transforming shell kinetic energy into hot-spot pressure, but they are also more sensitive to the deleterious effects of hydrodynamic instabilities and lead to lower energy gains based on their small mass. It follows that the massive slow shells (with equal mass and kinetic energy) yield much lower peak hot-spot pressures as compared to fast thin shells, but are less sensitive to hydrodynamic instabilities. The reduction in hot-spot pressures at low implosion velocities¹⁶ prevents the achievement of ignition in conventional ICF, but because of their good hydrodynamic stability properties and their potential for high energy gains, it is worth investigating possible mechanisms to transfer energy from the shell into the hot spot in thick, low-velocity shells.

This paper explains how shocks can be used to maximize the conversion of shell kinetic energy to internal energy in the hot spot and augment the compression of the central hot spot above the ignition threshold. The remainder of the paper is divided into four sections: Sec. II is dedicated to understanding planar implosions and the parameters that influence implosion performance. It includes analytic estimates of the theoretical maximum hot-spot pressure and scaling laws based on simulated target performances. Section III presents different ignitor shock techniques and compares implosions augmented with and without shocks. Sections IV and V discuss the implications of these techniques.

II. PLANAR IMPLOSIONS

For simplicity, ICF implosions are simulated starting near the end of the free-fall phase, when the laser driver has shut off and the main compression shock has reached the center of the target as shown in Fig. 1(a). We assume that all of the laser-plasma interactions have taken place and we are left with a system that depends only on the target geometry and composition and ignores heat transport, burn, and ablation physics to simplify the implosion to a system that is intuitive and easy to model while still capturing the main features of an ICF implosion. If we convert the implosion from spherical geometry to planar geometry, we can eliminate the effects of convergence which complicate the physical picture. In this way, we have simplified the complex dynamics of ICF implosions to be a simple compressible piston-like problem as shown in Fig. 1(b). The simulation's initial condition is a high-density slab compressing a low-density slab against a rigid wall, where the high- and low-density slabs correspond to the target shell and hot spot,

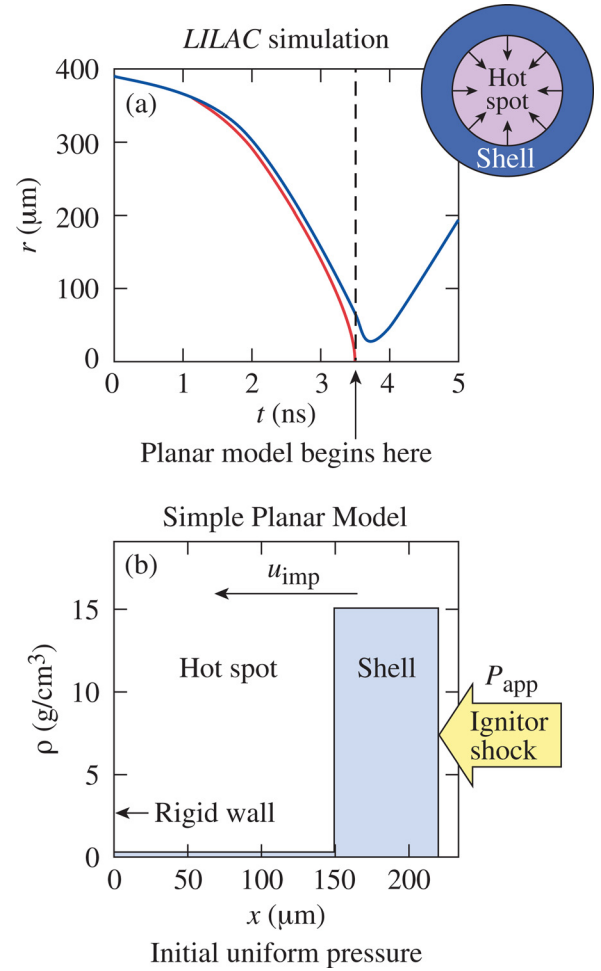


FIG. 1. (Color online) (a) Drawing of the shell's center of mass versus time in an ICF implosion. (b) The initial condition of the planar model simulates an ICF capsule during the coasting phase.

respectively. The pressure and velocity distributions are initially uniform, and pressure may be applied at the outer edge to simulate the ignitor shock.

A. Theoretical maximum hot-spot pressure

Using a 1-D planar Lagrangian hydrodynamic code, we can simulate an implosion from the end of its free-fall phase to stagnation. Figure 2 is the implosion pressure at the rigid wall, or the center of the hot spot, as a function of time for a 50- μm -thick slab with an initial uniform pressure of 50 Mbar, shell density of 15 g/cm^3 , and velocity of 3×10^7 cm/s , compressing a 150- μm lighter slab with a density of 0.3 g/cm^3 initially moving with the same velocity and at the same pressure. Notice that there are three distinct regions in the central pressure evolution in Fig. 2, illustrating three separate features in ICF implosions. The first region ($0 < t < t_1$) demonstrates the rise in pressure as shocks are reflected from the rigid wall, while the second ($t_1 < t < t_2$) and third ($t > t_2$) regions show the effect of rarefaction waves decompressing the assembly.

We can model the first region of the implosion analytically using the Rankine–Hugoniot (RH) relations¹⁷ to trace the path of any shocks and calculate the post-shock state of

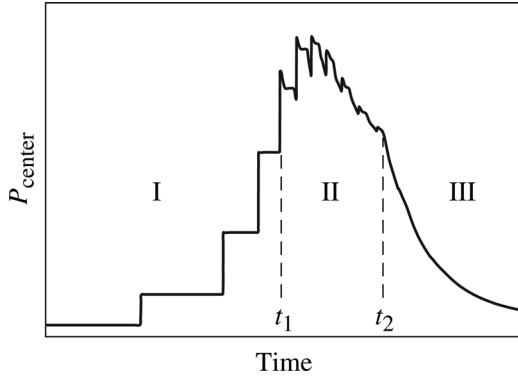


FIG. 2. Pressure at the rigid wall as a function of time for a planar compressible piston-like target.

the slabs. If we consider γ to be constant, the RH relations between the pre-shock (subscript i) and post-shock (subscript j) states in the frame of the moving shock are

$$z_{ji} = \frac{p_j - p_i}{p_i}, \quad (1)$$

$$M_{ji} = \sqrt{1 + \frac{\gamma + 1}{2\gamma} z_{ji}}, \quad (2)$$

$$\frac{u_j - u_i}{a_i} = \frac{z_{ji}}{\gamma M_{ji}}, \quad (3)$$

$$\frac{\rho_j}{\rho_i} = \frac{M_{ji}^2}{1 + \frac{\gamma - 1}{2\gamma} z_{ji}}, \quad (4)$$

$$\frac{a_j}{a_i} = \sqrt{\frac{(1 + z_{ji}) \left(1 + \frac{\gamma - 1}{2\gamma} z_{ji}\right)}{M_{ji}^2}}, \quad (5)$$

where p is the pressure, z is the shock strength, M is the Mach number of the shock, u is the material velocity, ρ is the material density, and a is the material sound speed ($a = \sqrt{\gamma p / \rho}$). By transferring to the lab frame, one can calculate the velocity of the shock (U) via $u = U - aM$.

At $t = 0$, the low-density slab encounters the rigid wall and a center-reflected shock wave is created that travels from the rigid wall to the high-density region as shown in Fig. 3(a). In this figure and in the following sketches, similar to Fig. 3, the shaded region represents the high-density region and the unshaded region represents the low-density region. The center-reflected shock can be calculated exactly by setting the velocity in the post-shocked region (u_1) to zero and solving the RH relations for the post-shock pressure,

$$\frac{p_3}{p_1} = 1 + \frac{\gamma u_1^2}{4a_1^2} (\gamma + 1) \left\{ 1 - \sqrt{1 + \left[\frac{4a_1}{u_1(\gamma + 1)} \right]^2} \right\}, \quad (6)$$

where p_k , u_k , and a_k , are the pressure, material velocity, and material sound speed, respectively, in region k . This can then be substituted into Eqs. (1)–(5) to solve for the rest of the unknown parameters. Once the center-reflected shock reaches the density discontinuity at the interface between the two den-

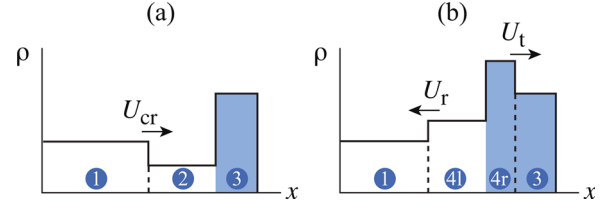


FIG. 3. (Color online) Sketch of (a) a center reflection with an outward-propagating shock (U_{cr}) and (b) a post-shock-density discontinuity collision that results in a reflected (U_r) shock and a transmitted (U_t) shock, which propagate in opposite directions.

sities, a reflected shock will propagate into the lower-density region (U_r) and a transmitted shock will propagate into the higher-density region (U_t), as shown in Fig. 3(b). It is important to note that across the density discontinuity, a pressure discontinuity may not exist as well any mass flux, stipulating that the pressure and material velocity across the entire post-shocked region must be uniform. Combining RH relations (1) and (3), the following equality results:

$$u_3 - \frac{\frac{a_3}{\gamma} \left(\frac{p_4}{p_3} - 1 \right)}{\sqrt{1 + \frac{\gamma + 1}{2\gamma} \left(\frac{p_4}{p_3} - 1 \right)}} = -u_2 + \frac{\frac{a_2}{\gamma} \left(\frac{p_4}{p_2} - 1 \right)}{\sqrt{1 + \frac{\gamma + 1}{2\gamma} \left(\frac{p_4}{p_2} - 1 \right)}}. \quad (7)$$

A numerical solution is used at this point to solve the sixth-order polynomial for the post-reflected and post-transmitted pressures (p_4 , in regions 4l and 4r, respectively). The reflected shock wave will propagate back through the low-density region and will reach the rigid wall, generating a second center-reflected shock wave. A comparison to the numerical simulation is shown in Fig. 4. Each rise in pressure, or “step,” in the pressure evolution corresponds to a new center reflection. The analytic calculation and the numerical simulation agree in both the value and timing of the pressure evolution up until $t = t_1$. At this point, the numerical simulation predicts a rarefaction wave reaching the center and decompressing the target. The origin of this rarefaction wave will be discussed in Subsection II B. The analytic calculation has ignored the generation of rarefaction waves and tracks the shocks only as

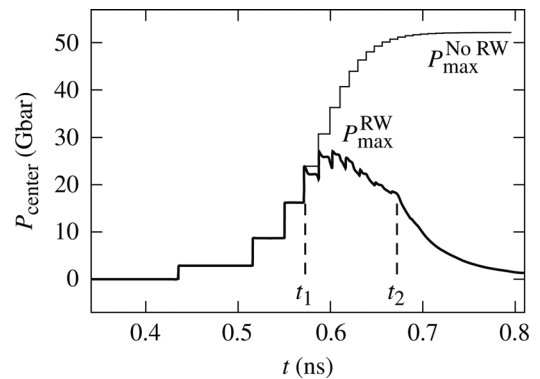


FIG. 4. Central pressure evolution as a function of time for a planar implosion. The thin line indicates analytic pressure evolution leading to a maximum pressure $P_{\max}^{\text{No RW}}$. The thick line indicates a numerical simulation of the same target, giving a maximum pressure $P_{\max}^{\text{RW}}$, including rarefaction waves.

they bounce between the rigid wall and the density discontinuity. The shock will continue to reflect off the rigid wall and the shell indefinitely, and the central pressure will approach a limiting value ($P_{\max}^{\text{noRW}}$) as shown in Fig. 4. This maximum pressure is the theoretical maximum central pressure that can be attained in the absence of rarefaction waves.

A simple dimensional analysis argument shows that the ratio of the maximum pressure to the initial pressure can depend only on dimensionless ratios of other values used to define the system: the ratios of the initial implosion velocity to the material sound speed, the shell to gas densities, and the shell's outer to inner thickness, i.e., $P_{\max}/P_0 = (u_{\text{imp}}/c_s)^{x_1} (\rho_{\text{shell}}/\rho_{\text{gas}})^{x_2}$. For simplicity, each target will be considered to be “thick,” so that the scaling laws may omit the shell aspect ratio as a variable. The reason for doing so will also be explained in Subsection II B. Nonlinear regression of the results of several calculations of varying initial conditions that simulate realistic free-fall conditions, $1 < p_0^{\text{(Mbar)}} < 100$, $0.15 < \rho_{\text{gas}}^{\text{(g/cm}^3\text{)}} < 37.5$, $7.5 < \rho_{\text{shell}}^{\text{(g/cm}^3\text{)}} < 37.5$, $1 < u_0^{\text{(10}^7\text{cm/s)}} < 4$, yields the following scaling law for the maximum pressure for a target with a 50- μm -thick shell and a 150- μm inner radius:

$$P_{\max}^{\text{noRW}} \approx \frac{1}{4} \rho_{\text{shell}} u_0^2 \left(\frac{\rho_{\text{shell}}}{\rho_{\text{gas}}} \right)^{0.7}. \quad (8)$$

The maximum pressure scales as the initial shell kinetic energy multiplied by a function of the ratio of the shell and gas densities. Dividing Eq. (8) by the initial pressure shows that the peak central pressure scales as the Mach number squared, comparable to the results of Ref. 16.

B. The effect of rarefaction waves

To understand the pressure evolution in regions II and III of Fig. 2, shock dynamics in the shell must be considered. Rarefaction waves bring information to the rest of the assembly relating to the finite compressibility of the slabs, essentially decompressing the target. For simplicity, we group the rarefaction waves into two types: rarefaction waves generated in the lower- or higher-density slab due to shocks interacting with density or pressure discontinuities (type 1) and rarefaction waves that have been generated on the outer edge of the high-density slab (type 2). Type 1 rarefaction waves decompress the assembly to specific densities and pressures and correspond to region II in Fig. 2, while type 2 rarefaction waves will decompress the assembly completely and corre-

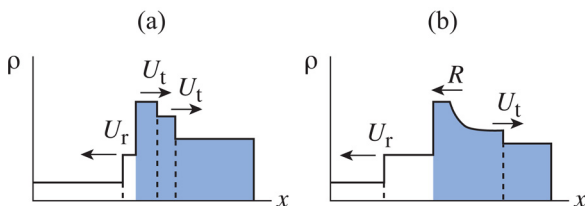


FIG. 5. (Color online) Sketch of the generation of type 1 rarefaction waves: (a) before the merging of two transmitted shocks (U_t) and (b) post-shock merge with the resulting transmitted shock (U_t) and reflected rarefaction wave (R).

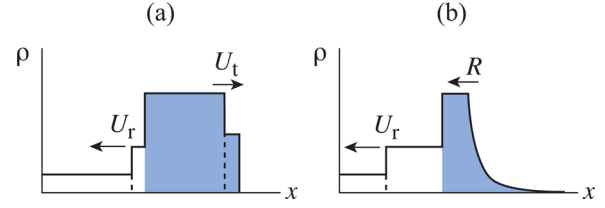


FIG. 6. (Color online) Sketch of the generation of type 2 rarefaction waves (a) before a transmitted shock (U_t) reaches the outer shell edge and (b) post-shock outer shell edge's interaction with the resulting rarefaction wave propagating inward (R).

spond to region III. Figures 5 and 6 give a sketch of how type 1 and type 2 rarefaction waves are generated.

Measuring the full effect of rarefaction waves on the maximum pressure requires numerical simulations because analytically modeling multiple shock interactions with density gradients is very complicated. After the maximum central pressure has been reached, numerical simulations show that the central pressure levels off and later decays as the slab expands. In thick shells (like this example), the maximum central pressure can be reached before type 2 rarefaction waves are launched back toward the hot spot. The arrival of this rarefaction wave can be seen by the abrupt change in the decay rate of the center pressure at $t \sim t_2$ (Fig. 2) and is the boundary between regions II and III. Running multiple numerical simulations with the same initial conditions as before will find the scaling law for the maximum effective pressure achievable with type 1 rarefaction waves,

$$P_{\max}^{\text{RW}} \approx 1.04 \rho_{\text{shell}} u_0^2 \left(\frac{\rho_{\text{shell}}}{\rho_{\text{gas}}} \right)^{0.19}. \quad (9)$$

The previous implosions were restricted to “thick shells,” where type 2 rarefaction waves did not affect the maximum central pressure. This simplified the implosions and allowed us to develop a scaling law, but does not encompass all ICF implosions. Figure 7 illustrates multiple implosions where the shell thickness is varied to illustrate the effect of type 2 rarefaction waves. The central pressure increases with increasing shell thickness until a critical thickness is reached (Δ_{crit}) where the central pressure plateaus. This is due to type 2 rarefaction waves decompressing the target before it has reached its maximum effective pressure. Once the shell thickness is beyond the critical thickness, the limiting factor to the maximum pressure is due to the effect of type 1 rarefaction waves.

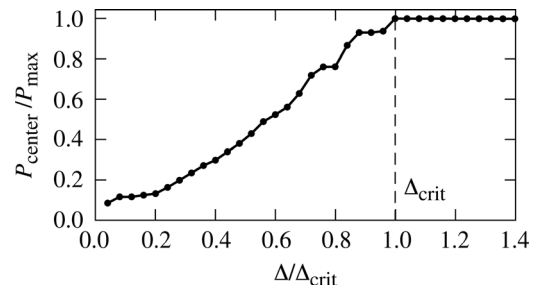


FIG. 7. Normalized central pressure versus normalized shell thickness. This indicates the central pressure is only a function of shell thickness for thin targets.

Therefore, for any $\Delta_{\text{shell}} > \Delta_{\text{crit}}$, the maximum central pressure is independent of the shell thickness. This leads to the conclusion that increasing the kinetic energy by increasing the shell thickness (with constant implosion velocity) does not raise the peak central pressure. Any additional mass, and therefore kinetic energy, that is added does not contribute to raising the hot-spot pressure. This is the reason for omitting the aspect ratio in the scaling laws.

III. PLANAR SHOCK IMPLOSIONS

The idea behind SI is to redistribute the energy in the laser pulse to drive a late shock and produce a non-isobaric fuel assembly that results in a significant reduction in the ignition threshold ($E_{\text{ign}} \sim P_{\text{hs}}^{-3}$) (Refs. 15 and 16). By reducing the implosion velocity of the target and using this energy instead to drive an ignitor shock, one can convert more shell kinetic energy into hot spot internal energy while preserving the attractive hydrostability properties of thick shells. This section will present three shock techniques aimed at mitigating the effect of rarefaction waves and maximizing the hot-spot pressure. Note that the amount of ignitor shock energy required is calculated in planar geometry by integrating the conservation of energy equation over the mass affected by the shock over all time. The energy of the pulse per unit area of the shock pulse is $E_{\text{pulse}} = P_{\text{ig}} u(P_{\text{ig}}) \Delta t$, where P_{ig} is the pressure of the pulse, $u(P_{\text{ig}})$ is the material velocity of the shell in the post-shock state, and Δt is the length of time when the pressure pulse is on.

A. No-rarefaction technique

Section II identified type 1 rarefaction waves as an important inhibiting factor on implosion performance. An implosion like the one shown in Fig. 4 can use weak shocks to eliminate type 1 rarefaction waves and “push” t_1 to t_2 . By sending a weak shock of the correct strength to collide with the first two transmitted shocks exactly at the time they merge, the result from the triple shock collision will be a single transmitted shock traveling to the shell outer edge and no shocks or rarefaction waves traveling inward. The result is a unique solution to the RH relations and is sketched in Fig. 8. This technique delays the time the hot spot receives information on the finite compressibility of the shell, reducing the duration of region II and increasing the maximum effective central pressure. The correct shock strength to apply for this technique may be found by setting the triple-shock-collision’s inward propagating shock wave to a shock strength of zero and solving the corresponding RH relations.

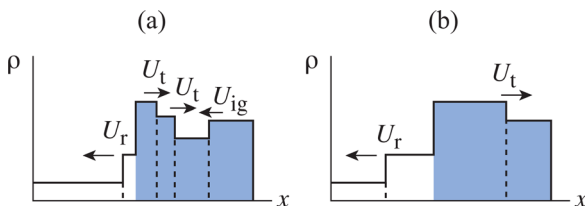


FIG. 8. (Color online) Sketch of the no-rarefaction technique: (a) before the ignitor shock (U_{ig}) and the first two transmitted shocks merge (U_t) and (b) after the three shocks collide, resulting with a single shock propagating outward (U_r).

This technique was simulated with a 50- μm -thick slab having an initial uniform pressure at 50 Mbar and a density of 15 g/cm³, compressing a 200- μm -thick low-density region with a density of 0.3 g/cm³ moving with the same velocity. The dashed curve in Fig. 9 is the central pressure versus time for an implosion with an initial velocity of 3.105×10^7 cm/s. The maximum central pressure for this case reaches ~ 28 Gbar. The solid curve in Fig. 9 demonstrates the no-rarefaction technique. In this case, a target of the same dimensions but with a reduced implosion velocity (2.98×10^7 cm/s) is augmented with multiple weak shocks. The extra energy attained from the reduction in the target’s kinetic energy is instead used to drive a 280-ps weak shock at 69 Mbar beginning 55 ps after the implosion, a second weak shock at 86 Mbar for 165 ps after the first weak shock, and a third weak shock at 97 Mbar for 309 ps. Therefore, for the same driver energy, the no-rarefaction technique makes it possible for the shocked target to follow the theoretical pressure staircase of the un-shocked no-rarefaction case and boost its maximum pressure to ~ 36 Gbar.

Although this case used only three weak shocks, multiple weak shocks may be launched such that multiple rarefaction waves can be suppressed. In theory, all of the rarefaction waves can be suppressed using this approach and the second region of the implosion can be eliminated completely, leading the pressure evolution to approach the theoretical limit in the absence of type 2 rarefaction waves. The maximum pressure amplification with infinite weak shocks can be found by dividing Eq. (8) by (9),

$$\frac{P_{\text{max}}^{\text{noRW}}}{P_{\text{max}}^{\text{RW}}} = \frac{1}{4} \left(\frac{\rho_{\text{shell}}}{\rho_{\text{gas}}} \right)^{0.5} \approx 1.77. \quad (10)$$

Although the application of the (three-pulse) no-rarefaction technique increased the maximum pressure by $\sim 30\%$, the disadvantage to this technique is being able to experimentally control these weak shocks to such a high precision. The shock timing and strength must be controlled to such a minute value that any perturbations in the shock uniformity would likely negate any experimental benefit. Additionally, even if all of the type 1 rarefaction waves were eliminated, a transmitted shock would eventually reach the shell’s outer surface and generate a type 2 rarefaction wave that would

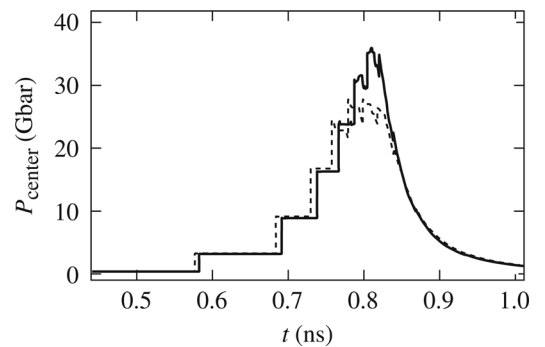


FIG. 9. Pressure (Gbar) at the rigid wall as a function of time (ns), comparing numerical simulations of a no-rarefaction target (solid line) and a non-shocked target (dashed line). The temporal offset is due to a difference in implosion velocities to ensure both targets have the same total energy.

decompress the target before the analytic limit without rarefaction waves could be reached.

B. No-transmission technique

The generation of type 2 rarefaction waves is inevitable in ICF implosions but may be delayed by taking advantage of another unique RH relation solution. By timing a shock of the correct strength so that it encounters the density discontinuity just as the first center-reflected shock does, one can “cancel” the transmitted shock. The result of the double-shock, density-discontinuity collision is a single reflected shock wave traveling inward. This technique is similar to the previously demonstrated no-rarefaction technique (cf. Sec. III A) and is illustrated in Fig. 10. This affects the implosion in multiple ways: (1) once the reflected shock wave reaches the center, the implosion is virtually reset, i.e., the implosion is similar to the initial condition but with a larger pressure, density, and velocity; (2) canceling the first transmitted shock wave delays the generation of a type 2 rarefaction wave; and (3) this technique takes advantage of the natural resonance of the hot-spot shocks. Similar to a harmonic oscillator driven at its natural frequency, the hot spot of the assembly is driven at its “natural frequency,” which varies with time in order to achieve maximum compression.

This technique was simulated with a 50- μm -thick slab with an initial uniform pressure at 50 Mbar and a density of 15 g/cm³ moving at a velocity of 2.6×10^7 cm/s, compressing a 150- μm -thick, 0.3 g/cm³ slab initially moving with the same velocity. It has the same kinetic energy (69 kJ/mm²) as a target of the same dimensions but an initial velocity of 4.1×10^7 cm/s. Approximately 1 ps after the start of the simulation, a 517 ps, 2.1 Gbar ignitor shock is driven on the outer edge and propagates inward, colliding with the first center-reflection shock wave at the gas-shell interface. A single reflected shock propagates back toward the center, leaving an ~ 13.5 - μm -thick slab with a uniform pressure of 2.1 Gbar and a shell density of 55 g/cm³ moving at a velocity of 3.6×10^7 cm/s, compressing an 18- μm -thick, low-density region of 2.0 g/cm³ moving at the same velocity. The new implosion obeys the same hydrodynamics as before, and therefore, we are able to use the scaling relation for the maximum pressure [Eq. (9)] for the “reset” implosion as well. Taking the ratio of the scaling laws for a shocked implosion and an unshocked implosion gives the theoretical pressure amplification,

$$\frac{P_{\text{shock}}}{P_{\text{no shock}}} = \frac{\rho'_{\text{shock}} u_0'^2}{\rho_{\text{shell}} u_0^2} \left(\frac{\rho'_{\text{shell}} / \rho'_{\text{gas}}}{\rho_{\text{shell}} / \rho_{\text{gas}}} \right)^{0.19}, \quad (11)$$

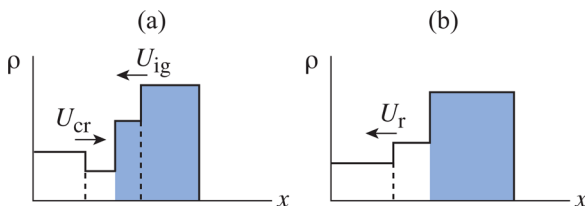


FIG. 10. (Color online) Sketch of the no-transmission technique: (a) before the center-reflected shock (U_{cr}) and the ignitor shock (U_{ig}) collide; (b) post-collision with a reflected shock propagating back to the rigid wall (U_r).

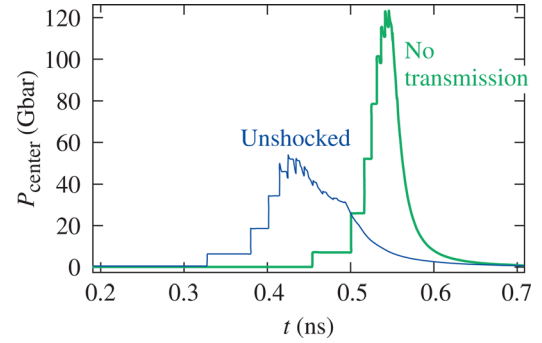


FIG. 11. (Color online) Pressure (Gbar) at the rigid wall as a function of time (ns) comparing numerical simulations of an unshocked target (thin blue line) and a no-transmission target (thick green line) of equal implosion energy and target geometry.

where the primes indicate the shocked values. This technique can, therefore, theoretically amplify the maximum central pressure by a factor of ~ 2.5 for this specific target. The maximum effective pressure for the unshocked case is ~ 54 Gbar versus the shocked case of ~ 123 Gbar, in agreement with Eq. (11). The pressure evolution profiles are displayed in Fig. 11.

C. Re-shock technique

Because the no-transmission technique requires a long ignitor pulse in order to achieve its pressure amplification, a sizeable amount of energy must be devoted to maintain the pulse. If instead a shorter, more-intense ignitor pressure pulse is used, it is possible to launch it closer to stagnation and allows for the target to assemble to large densities and pressures before shocking the hot spot again. Since the central pressure is only affected by additional shocks and rarefaction waves, the ideal time to launch this shock would be to time it in the same manner as the no-rarefaction technique. The difference, however, is to set the pressure of the ignitor shock's strength equal to that of the second center reflection's transmitted shock at the merging time, so that the collision will produce a reflected and transmitted shock, as shown in Fig. 12. By doing so, the rarefaction wave may be replaced with a shock wave. This increases the number of pressure steps in the hot spot in the early stages of the hot-spot compression and delays the time at which the information of the finite shell compressibility reaches the hot spot. This technique is similar to the original ignitor shock-launching time proposed by Betti *et al.*,¹ and is demonstrated with the same unshocked target as in Sec. II. This time, however, the implosion velocity of the target is reduced from 4.1×10^7 cm/s to 2.9×10^7 cm/s, and an ignitor pressure

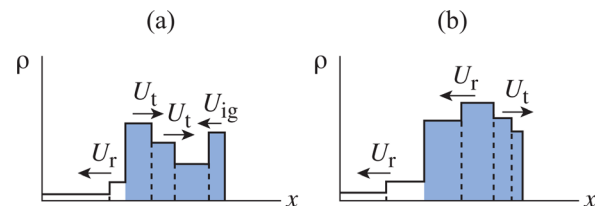


FIG. 12. (Color online) Sketch of the re-shock technique: (a) before the ignitor shock (U_{ig}) and the first two transmitted shocks (U_t) merge; (b) post-collision resulting in shocks propagating in opposite directions (U_l and U_r).

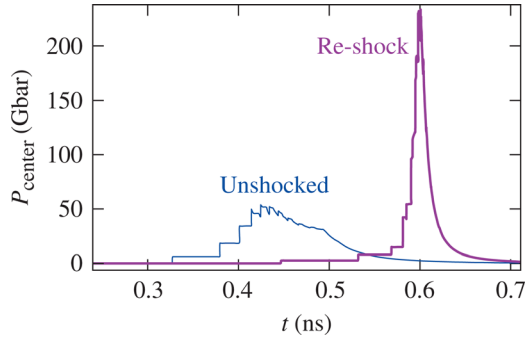


FIG. 13. (Color online) Pressure (Gbar) at the rigid wall as a function of time (ns) comparing numerical simulations of an unshocked target (thin blue line) and a re-shocked target (thick purple line) of equal implosion energy.

pulse of 5.2 Gbar that lasts for 142 ps is applied after 440 ps from the start of the simulation. The total energy (kinetic + ignitor shock) is 69 kJ/mm²—the same as the unshocked case. As shown in Fig. 13, the center pressure has increased to ~ 233 Gbar, over $4\times$ larger than its energy-equivalent unshocked case.

IV. RESULTS

All three of the previous ignitor shock techniques were focused on mitigating the effects of the rarefaction waves and were shown to perform better than their energy equivalent, but unshocked, targets. An equal energy comparison is shown in Fig. 14 over a range of total kinetic energy that illustrates the overall effectiveness of each technique. The dashed line indicates an extrapolated prediction for the no-transmission technique; any implosion in this energy regime needs the ignitor shock to be launched prior to the beginning of the simulation, i.e., before the end of the coasting phase. For a given total energy ($E_{\text{total}} = E_{\text{kinetic}} + E_{\text{shock}}$), the ignitor-pulse pressure was chosen based on the technique being used for a given v_{imp} and the pulse length was optimized to obtain the maximum central pressure with the least amount of pulse energy. Although Fig. 14 clearly shows that the re-shock technique performs best for all energies of interest, it should not, however, be considered the optimum technique for every scenario. For example, as the total energy scales up, the ignitor shock pressure approximately doubles the no-

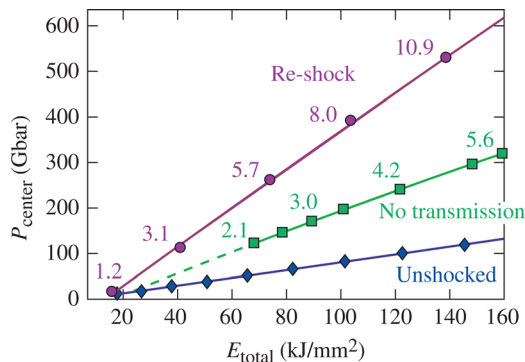


FIG. 14. (Color online) Simulated central pressure (Gbar) versus total energy (kJ) for energy- and mass-equivalent targets using different implosion techniques. Numbers indicate the required ignitor pulse pressure in Gbar.

transmission technique. This could lead high-energy targets to use the no-transmission technique if the driver is unable to generate large ignitor pressure pulses or if the laser intensity is limited by laser-plasma instabilities. Achieving ignitor pulse pressures of the order of several Gbar may be overly optimistic for laser-generated shocks; however, realistic implosions will benefit from convergence effects in spherical geometries that lead to more-realizable ignitor pulse pressures.

Because the implosions given in Fig. 14 scale approximately linearly to the total energy, one can find a relation between the total energy required (E_T) to drive the implosion and the energy that must be used in the ignitor pulse (E_P). If the total energy required to drive the implosion is $E_T = E_P + E_C$, where E_C is the reduced kinetic energy and is a function of the reduced implosion velocity given by Fig. 14, the ratio of the pulse energy required to the total energy is then given by

$$\frac{E_P}{E_T} = \frac{E_P}{E_P + E_C}. \quad (12)$$

This ratio will yield the percentage of on-target energy that must be used to drive the ignitor pulse compared to the total energy. Figure 15 plots Eq. (12) as a function of velocity for both the no-transmission and re-shock techniques. This points out that roughly 50% of the total drive energy must be put into the ignitor pulse in re-shock targets and $\sim 60\%$ into the no-transmission targets. Again it is important to note that by switching to a spherical geometry, the convergence effects will lower the amount of energy required in the ignitor pulse. A comprehensive study in spherical geometry completed by Schmitt *et al.*⁷ found the energy ratio to be approximately 20%. A pulse optimization was also carried out by Lafon *et al.*,⁹ where an optimum implosion velocity (of about 250 km/s) and optimum late shock timing, pulse energy, and intensity were found for realistic inertial fusion energy targets. Therefore, given a specific technique, total energy, and capsule geometry, the peak implosion velocity, ignitor shock strength, and the energy ratio between the initial energy and the ignitor pulse energy for the optimal implosion are uniquely specified.

Although this paper is mostly concerned with the application of single ignitor shocks, it is possible to use the different techniques presented in sequential fashion to further

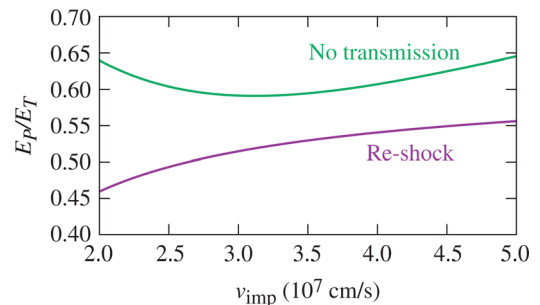


FIG. 15. (Color online) Ratio of the ignitor pulse energy to total energy required for no-transmission and re-shock techniques as a function of reduced implosion velocity.

enhance the implosion performance. Like the no-rarefaction technique, the no-transmission technique can also be applied multiple times to the same target. Given the no-transmission example in Sec. III B, a second no-transmission shock can be launched ~ 433 ps later at 29 Gbar to “reset” the implosion once again. The resulting implosion generates a hot spot pressure of 850 Gbar, but requires 178 kJ/mm^2 of kinetic driver energy. Additionally, the re-shock technique can also be used in conjunction with the no-transmission technique. For example, if the re-shock technique is applied after the same example in Sec. III B, the resulting implosion can produce a maximum central pressure of 742 Gbar. In this case, the re-shock pressure pulse is 37 Gbar and is applied ~ 503 ps after the 2 Gbar no-transmission shock and requires 121 kJ/mm^2 of total kinetic-driver energy. Therefore, although it may be beneficial for high-energy implosions to use multiple ignitor shocks, it may not be physically practical given current driver technology.

Another item to be considered is the shell’s critical thickness and its dependence on the ignitor shock. Not only does the normalized critical thickness vary with the same variables as the previously derived scaling laws [Eqs. (8) and (9)], it is also dependent on the strength and timing of the ignitor shock. This leads to another optimization parameter and would significantly improve the performance of the implosions where the shock has changed the critical thickness to be larger than the shell thickness. For example, the reduction in the maximum central pressure caused by $\Delta_{\text{shell}} < \Delta_{\text{crit}}$ (cf. Sec. II B) occurs in the lower-energy implosions of Fig. 14. These implosions could perform better if the ignitor-pulse energy was instead exchanged for more shell mass (and kinetic energy). A scaling law for the critical thickness and its dependence on the techniques presented will be investigated in a forthcoming paper.

V. CONCLUSIONS

This paper has presented some of the underlying physics behind shock ignition in an attempt to optimize the energy distribution in the shell and hot spot. The analysis of unshocked targets showed the main inhibitor to implosion performance (in terms of pressure) to be rarefaction waves created within the shell of the target which decompressed the capsule before the theoretical peak pressure could be reached. Three different techniques (no-rarefaction, no-transmission, and re-shock) aimed at mitigating the effect of these rarefaction waves in implosions augmented with shocks were presented, and all three were shown to perform better than their energy-equivalent unshocked targets. At best, by eliminating the rarefaction waves, the pressure could be increased by $\sim 80\%$, where the re-shock technique amplified the pressure by a factor of ~ 4 . Therefore, the performance

enhancement was also due to other factors, including the shock’s ability to effectively enhance the coupling of the shell’s kinetic energy to the hot spot’s internal energy by adjusting the critical thickness of the shell. Shock ignition, therefore, makes it possible for thick, hydrodynamically stable targets to gain the benefits of thin shells by reducing the effect of rarefaction waves and maximizing the conversion between the shell’s kinetic energy and the hot spot’s internal energy to achieve ignition with potentially high gains.

ACKNOWLEDGMENTS

The authors thank M.A. Roe for his helpful contributions. This work was supported by the U.S. Department of Energy Office of Inertial Confinement Fusion under Cooperative Agreement Nos. DE-FC52-08NA28302 and DE-FC02-04ER54789.

- ¹R. Betti, C. D. Zhou, K. S. Anderson, L. J. Perkins, W. Theobald, and A. A. Solodov, *Phys. Rev. Lett.* **98**, 155001 (2007).
- ²X. Ribeyre, G. Schurtz, M. Lafon, S. Galera, and S. Weber, *Plasma Phys. Controlled Fusion* **51**, 015013 (2009).
- ³A. J. Schmitt, J. W. Bates, S. P. Obenschain, S. T. Zalesak, D. E. Fyfe, and R. Betti, *Fusion Sci. Technol.* **56**, 377 (2009).
- ⁴L. J. Perkins, R. Betti, K. N. LaFortune, and W. H. Williams, *Phys. Rev. Lett.* **103**, 045004 (2009).
- ⁵X. Ribeyre, M. Lafon, G. Schurtz, M. Olazabal-Loumé, J. Breil, S. Galera, and S. Weber, *Plasma Phys. Controlled Fusion* **51**, 124030 (2009).
- ⁶S. Atzeni, J. R. Davies, L. Hallo, J. J. Honrubia, P. H. Maire, M. Olazabal-Loume, J. L. Feugeas, X. Ribeyre, A. Schiavi, G. Schurtz, J. Breil, and P. Nicolai, *Nucl. Fusion* **49**, 055008 (2009).
- ⁷A. J. Schmitt, J. W. Bates, S. P. Obenschain, S. T. Zalesak, and D. E. Fyfe, *Phys. Plasmas* **17**, 042701 (2010).
- ⁸G. Schurtz, X. Ribeyre, and M. Lafon, *J. Phys.: Conf. Ser.* **244**, 022013 (2010).
- ⁹M. Lafon, X. Ribeyre, and G. Schurtz, *Phys. Plasmas* **17**, 052704 (2010).
- ¹⁰W. Theobald, R. Betti, C. Stoeckl, K. S. Anderson, J. A. Delettrez, V. Yu. Glebov, V. N. Goncharov, F. J. Marshall, D. N. Maywar, R. L. McCrory, D. D. Meyerhofer, P. B. Radha, T. C. Sangster, W. Seka, D. Shvarts, V. A. Smalyuk, A. A. Solodov, B. Yaakobi, C. D. Zhou, J. A. Frenje, C. K. Li, F. H. Séguin, R. D. Petrasso, and L. J. Perkins, *Phys. Plasmas* **15**, 056306 (2008).
- ¹¹W. Theobald, K. S. Anderson, R. Betti, R. S. Craxton, J. A. Delettrez, J. A. Frenje, V. Yu. Glebov, O. V. Gotchev, J. H. Kelly, C. K. Li, A. J. Mackinnon, F. J. Marshall, R. L. McCrory, D. D. Meyerhofer, J. F. Myatt, P. A. Norreys, P. M. Nilson, P. K. Patel, R. D. Petrasso, P. B. Radha, C. Ren, T. C. Sangster, W. Seka, V. A. Smalyuk, A. A. Solodov, R. B. Stephens, C. Stoeckl, and B. Yaakobi, *Plasma Phys. Controlled Fusion* **51**, 124052 (2009).
- ¹²E. I. Moses, *J. Phys.: Conf. Ser.* **112**, 012003 (2008).
- ¹³R. Betti, W. Theobald, C. D. Zhou, K. S. Anderson, P. W. McKenty, S. Skupsky, D. Shvarts, V. N. Goncharov, J. A. Delettrez, P. B. Radha, T. C. Sangster, C. Stoeckl, and D. D. Meyerhofer, *J. Phys.: Conf. Ser.* **112**, 022024 (2008).
- ¹⁴O. Klimo, S. Weber, V. T. Tikhonchuk, and J. Limpouch, *Plasma Phys. Controlled Fusion* **52**, 055013 (2010).
- ¹⁵M. C. Herrmann, M. Tabak, and J. D. Lindl, *Nucl. Fusion* **41**, 99 (2001).
- ¹⁶C. D. Zhou and R. Betti, *Phys. Plasmas* **14**, 072703 (2007).
- ¹⁷G. B. Whitham, *Linear and Nonlinear Waves*, Pure and Applied Mathematics (Wiley, New York, 1974), p. 143.